Sushil Thakur

Avalonia UI | WPF | .NET Desktop Developer | Real-Time GUI | Figma-to-UI Implementation

+91 6201941071 | sushilthakur9792@gmail.com | LinkedIn | GitHub (SUBHASHSUSHIL) | WhatsApp

Profile Summary

.NET Desktop Developer with 3+ years of experience building intuitive, real-time desktop applications using Avalonia UI and WPF/XAML. Most recently built Video Management System (VMS) products in the R&D team at CP Plus, including multi-threaded, real-time GUI for live camera monitoring with UI implemented directly from Figma design specs. Strong proficiency in MVVM, Clean Architecture, design patterns, SOLID principles, and OOP, with additional backend experience in ASP.NET Core, REST APIs, SQL Server, and PostgreSQL. Contributed to 27+ software projects across Smart City, Video Surveillance, IoT, and Enterprise domains, with a focus on UI/UX quality, system performance, and reliability.

Technical Skills

- **Languages:** C#, Python, SQL
- **Backend & Frameworks:** .NET 8/10+, ASP.NET Core, MVC, REST API, CQRS, Minimal API
- **Desktop Development:** WPF, Avalonia UI, MVVM Architecture, Dependency Injection, XAML, Custom Controls
- **Frontend:** Angular, Bootstrap, HTML5, CSS3, JavaScript, Figma
- **Databases & Messaging:** MSSQL, PostgreSQL, SQLite, MySQL, InfluxDB, RabbitMQ
- **Cloud & DevOps:** Microsoft Azure, Docker, Git, GitHub, GitLab, CI/CD, Jira, Postman
- **Architecture & Practices:** Clean Architecture, MVVM, CQRS, Design Patterns, SOLID Principles, OOP, SOA, Repository Pattern, Multithreading, Unit Testing, Agile
- **AI/LLM Integration:** Model Context Protocol (MCP), Retrieval-Augmented Generation (RAG)

Professional Experience

Software Engineer – R&D | CP Plus *Mar 2026 – Sep 2026*

Noida, India

Project 1: VMS (Video Management System)

- **Multi-App Desktop Architecture:** Built two role-based desktop applications (Admin + Operator) in Avalonia UI (.NET 10) using MVVM with CommunityToolkit source generators and DI-based service composition.
- **Figma-to-UI Implementation:** Translated Figma design specs into pixel-accurate Avalonia/AXAML UI screens across both applications, maintaining design consistency through a shared control library.
- **Live Camera Video Wall:** Developed a real-time, multi-threaded HMI-style video wall (infinite-scroll, multi-camera) with drag-and-drop placement, resizable grid cells, fullscreen mode, and live per-cell status overlays, keeping the UI thread responsive while decoding multiple concurrent video streams.
- **ONVIF Integration:** Implemented ONVIF protocol support for automated camera discovery and PTZ (pan-tilt-zoom) control across connected devices.
- **In-App Support Chatbot:** Building a RAG-based support chatbot within the VMS Admin app, retrieving from system documentation to answer user queries about configuration and troubleshooting.
- **Role-Based Access & User Management:** Built full CRUD UI for users and roles with permission-key-based RBAC, gating access and auto-navigation by assigned role.
- **Tech Stack:** C#, Avalonia UI, .NET 10, MVVM, SignalR, ONVIF, RAG, JWT authentication

Project 2: CP DMT (Device Management Tool)

- **Multi-Step Production Workflow UI:** Built a structured desktop workflow (Scan → Flash → Burn → Functional Test → Boot Test → Print Label) that reduced operator errors by enforcing sequential step completion.
- **Firmware Flashing & Device Discovery:** Designed a real-time network camera scanning panel and an interactive Flash panel with progress tracking, MD5 firmware verification, and cancelable async operations.
- **Serial & MAC Burn + Label Printing:** Built a Burn panel to write Serial Number/MAC Address to devices with label preview and direct printer integration.
- **Tech Stack:** C#, Avalonia UI, .NET, MVVM, async/await, JWT authentication

Software Engineer | Ajeevi Technologies Pvt Ltd *Mar 2024 – Feb 2026*
Noida, India

Project 1: ICCC-Smart City (Integrated Command & Control Center)

- **High-Scale IoT Data:** Engineered real-time GPS dashboards and backend modules processing live telemetry from 1,000+ IoT devices with sub-second reporting latency.
- **Modern API Architecture:** Designed high-performance backend services using .NET Core Minimal APIs, reducing boilerplate and improving endpoint execution speed.
- **Multi-Database Orchestration:** Integrated MSSQL/MySQL for relational data with InfluxDB for high-velocity time-series sensor data.
- **Optimization:** Built a reusable backend component library, cutting development time for new features by approximately 20%.

Project 2: Video Management System (VMS)

- **Architectural Implementation:** Applied CQRS to separate read/write operations, improving API responsiveness and data consistency.
- **Performance Tuning:** Conducted deep-level MSSQL optimization, reducing query execution time for large video metadata sets.

Project 3: Smart Parking Management System (SPMS)

- **Scalable Service Design:** Built RESTful APIs and modularized backend services enabling horizontal scaling as parking site density increases.
- **Real-Time Synchronization:** Optimized MySQL/InfluxDB queries to handle simultaneous occupancy updates from hundreds of concurrent sensors.
- **Admin Dashboard:** Built an Angular-based admin monitoring dashboard displaying real-time parking occupancy and sensor health status across all sites.

Software Engineer Intern | So Infotech Pvt Ltd *Jul 2023 – Dec 2023*
Noida, India

Project: Concert-eTicket-Booking (Enterprise Ticket Management System)

- **End-to-End Development:** Engineered the core ticket-booking engine using ASP.NET Core MVC and SQL Server, focusing on transaction integrity during peak booking windows.
- **API Integration:** Developed secure REST APIs and user authentication modules.

Education

Master of Computer Applications (MCA) — Mangalayatan University, Aligarh, UP | CGPA: 8.4

Certification

Microsoft Certified: Azure Developer Associate — Microsoft | Issued 2026